Modern Processor Design (III): Right here waiting

Hung-Wei Tseng

Recap: how does compiler implement if-else?

```
for(i=0;i<size;i++)
{
    if (data[i] % 2 != 0)
        call_when_true(&data[i]);
    else
        call_when_false(&data[i]);
}
return 0;
```

go to L10 when data[i] % 2 == 1

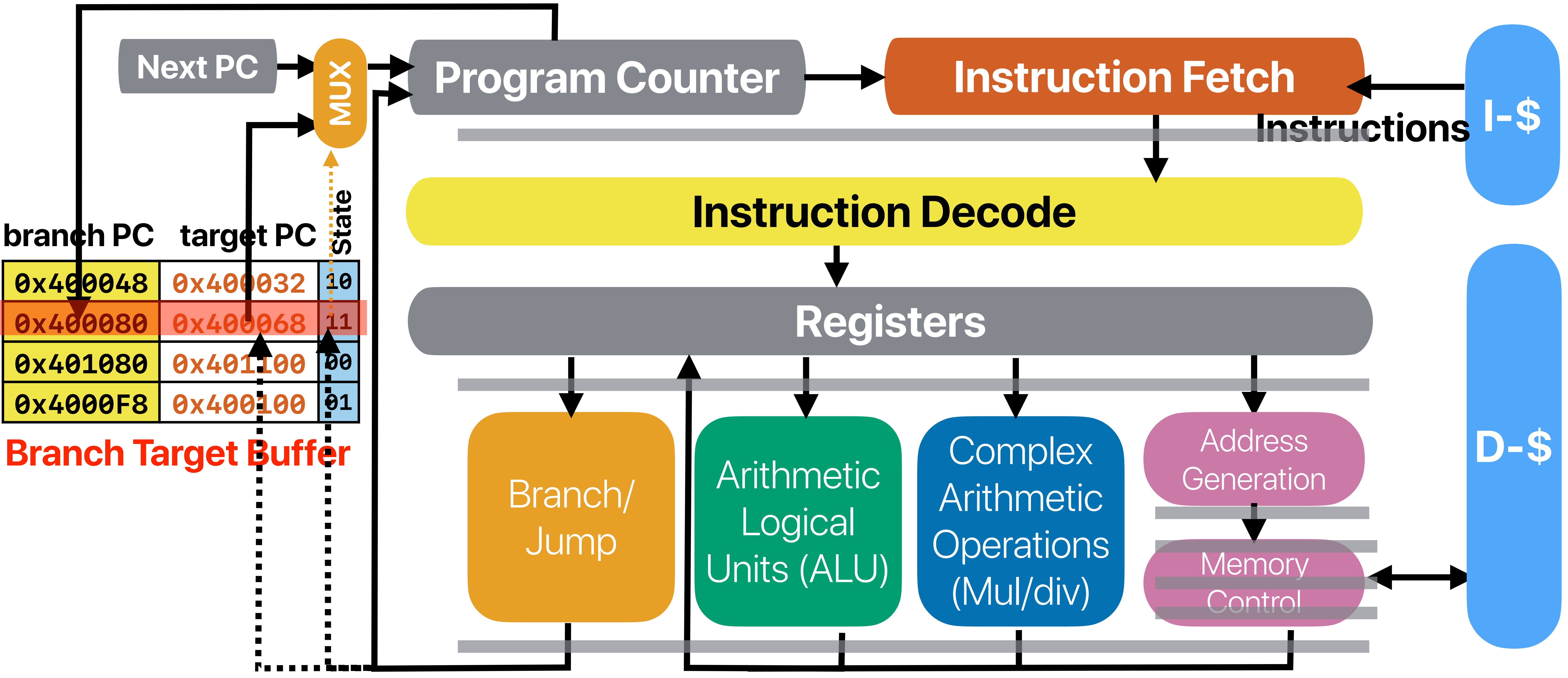
// branch taken if it's false

branch taken (changing address to L11) if the evaluation is not true

```
.L12:
    movl    -32(%rbp), %eax
    cltq
    leaq    0(,%rax,8), %rdx
    movq    -8(%rbp), %rax
    addq    %rdx, %rax
    movq    (%rax), %rax
    andl    $1, %eax
    testq   %rax, %rax
    je  .L10
    movl    -32(%rbp), %eax
    cltq
    leaq    0(,%rax,8), %rdx
    movq    -8(%rbp), %rax
    addq    %rdx, %rax
    movq    %rax, %rdi
    call    call_when_true
    jmp .L11
.L10:
    movl    -32(%rbp), %eax
    cltq
    leaq    0(,%rax,8), %rdx
    movq    -8(%rbp), %rax
    addq    %rdx, %rax
    movq    %rax, %rdi
    call    call_when_false
.L11:
    addl    $1, -32(%rbp)
.L9:
    movl    -32(%rbp), %eax
    cmpl    -28(%rbp), %eax
    jl  .L12
    movl    $0, %eax
    leave
```

In if-else statement, true can mean "not taken"

Detail of a basic dynamic branch predictor



2-bit local predictor

- What's the overall branch prediction (include both branches) accuracy for this nested for loop?

```
i = 0;
do {
    if( i % 2 != 0) // Branch X, taken if i % 2 == 0
        a[i] *= 2;
    a[i] += i;
} while ( ++i < 100)// Branch Y
```

Can we do a better job?

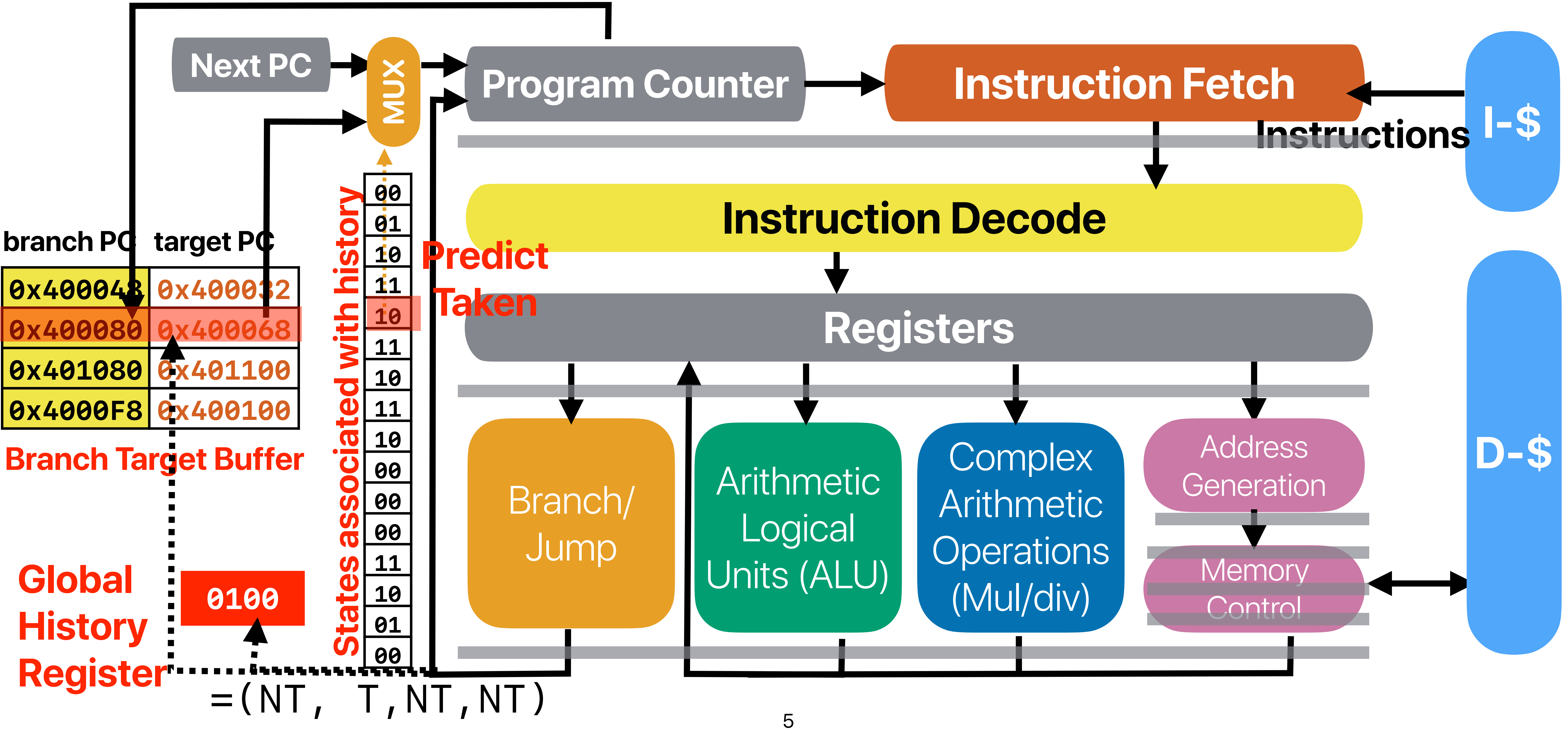
(assume all states started with 00)

A. ~25%
B. ~33%
C. ~50%
D. ~67%
E. ~75%

For branch Y, almost 100%, For branch X, only 50%

i	branch?	state	prediction	actual
0	X	00	NT	T
1	Y	00	NT	T
1	X	01	NT	NT
2	Y	01	NT	T
2	X	00	NT	T
3	Y	10	T	T
3	X	01	NT	NT
4	Y	11	T	T
4	X	00	NT	T
5	Y	11	T	T
5	X	01	NT	NT
6	Y	11	T	T
6	X	00	NT	T
7	Y	11	T	T

Detail of a basic dynamic branch predictor



Performance of GH predictor

```
i = 0;
do {
    if( i % 2 != 0) // Branch X, taken if i % 2 == 0
        a[i] *= 2;
    a[i] += i;
} while ( ++i < 100)// Branch Y
```

i	branch?	GHR	state	prediction	actual
0	X	000	00	NT	T
1	Y	001	00	NT	T
1	X	011	00	NT	NT
2	Y	110	00	NT	T
2	X	101	00	NT	T
3	Y	011	00	NT	T
3	X	111	00	NT	NT
4	Y	110	01	NT	T
4	X	101	01	NT	T
5	Y	011	01	NT	T
5	X	111	00	NT	NT
6	Y	110	10	T	T
6	X	101	10	T	T
7	Y	011	10	T	T
7	X	111	00	NT	NT
8	Y	110	11	T	T
8	X	101	11	T	T
9	Y	011	11	T	T
9	X	111	00	NT	NT
10	Y	110	11	T	T
10	X	101	11	T	T
11	Y	011	11	T	T

Near perfect after this

Demo revisited: evaluating the cost of mis-predicted branches

- Compare the number of mis-predictions
- Calculate the difference of cycles
- We can get the “average CPI” of a mis-prediction!

34 cycles on Intel Alder Lake

24 cycles on AMD Zen 3

Could be more expensive than cache misses

Better predictor?

- Consider two predictors — (L) 2-bit local predictor with unlimited BTB entries and (G) 4-bit global history with 2-bit predictors. How many of the following code snippet would allow (G) to outperform (L)?

I — about the same

```
i = 0;
do {
    if( i % 10 != 0)
       a[i] *= 2;
    a[i] += i;
} while ( ++i < 100);
```

II — about the same

```
i = 0;
do {
    a[i] += i;
} while ( ++i < 100);
```

III ✓

```
i = 0;
do {
    j = 0;
    do {
      sum += A[i*2+j];
    }
    while( ++j < 2);
} while ( ++i < 100);
```

IV — L could be better

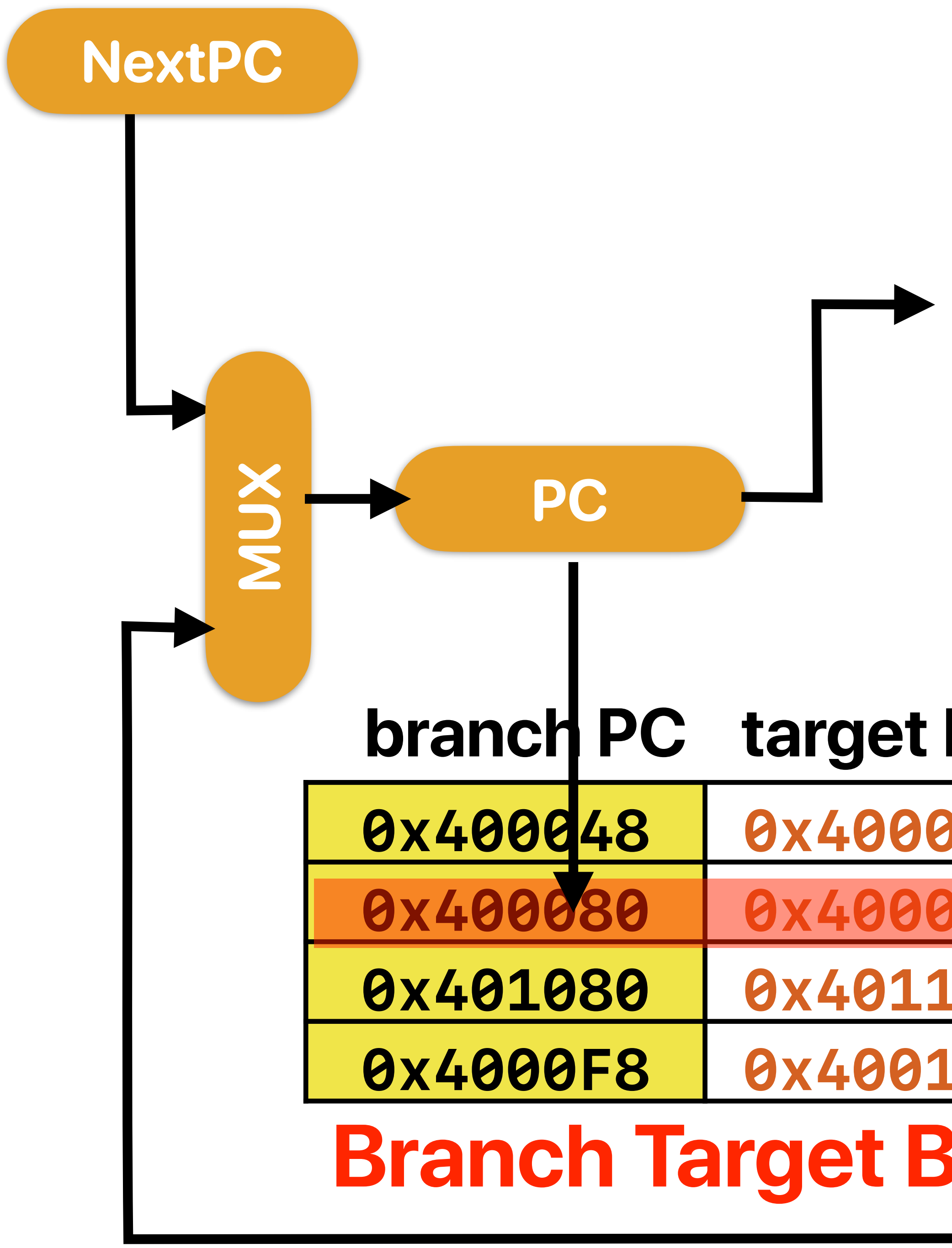
```
i = 0;
do {
    if( rand() %2 == 0)
       a[i] *= 2;
    a[i] += i;
} while ( ++i < 100)
```

A. 0
B. 1
C. 2
D. 3
E. 4

Takeaways: branch predictions

- The cost of not to predict a branch is to stall until the data dependency is resolved — 34 cycles on modern intel processors and 24 on AMD processors
- Branch predictions allow the processor to at least make some progress and hide the stalls if we guessed correctly!
- Dynamic branch prediction — predict based on prior history
    - Local predictor — make predictions based on the state of each branch instruction
    - Global predictor — make predictions based on the state from all branches
    - Both are not perfect — that's why we need hybrid predictors!

Tournament Predictor



branch PC	target PC	State
0x400048	0x400032	1
0x400080	0x400068	1
0x401080	0x401100	1
0x4000F8	0x400100	0

Branch Target Buffer

Global History Register

0100

States associated with history

00
01
10
11
10
11
10
11
10
00
00
00
11
10
01
00

Local History Predictor

branch PC	local history
0x400048	1000
0x400080	0110
0x401080	1010
0x4000F8	0110

Predict Taken

States associated with history

00
01
10
11
10
11
10
11
10
00
00
00
11
10
01
00

Outline

- Hybrid predictors (cont.)
- Data hazards
- Hardware optimizations for data hazards

Hybrid predictors (cont.)

Perceptron

Jiménez, Daniel, and Calvin Lin. "Dynamic branch prediction with perceptrons." Proceedings HPCA Seventh International Symposium on High-Performance Computer Architecture. IEEE, 2001.

The following slides are excerpted from https://www.jilp.org/cbp/Daniel-slides.PDF by Daniel Jiménez

Branch Prediction is Essentially an ML Problem

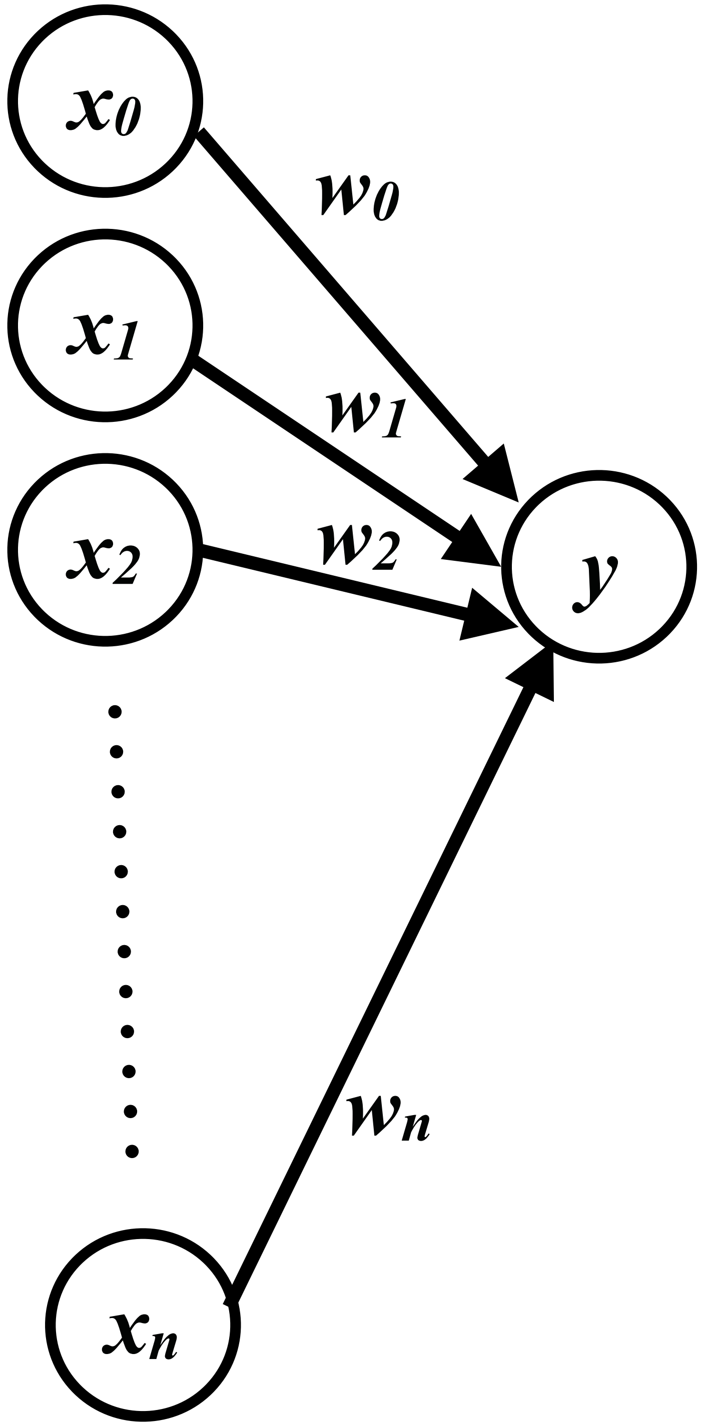
- The machine learns to predict conditional branches
- Artificial neural networks
  - Simple model of neural networks in brain cells
  - Learn to recognize and classify patterns

Mapping Branch Prediction to NN

- The inputs to the perceptron are branch outcome histories
  - Just like in 2-level adaptive branch prediction
  - Can be global or local (per-branch) or both (alloyed)
  - Conceptually, branch outcomes are represented as
    - +1, for taken
    - -1, for not taken
- The output of the perceptron is
  - Non-negative, if the branch is predicted taken
  - Negative, if the branch is predicted not taken
- Ideally, each static branch is allocated its own perceptron

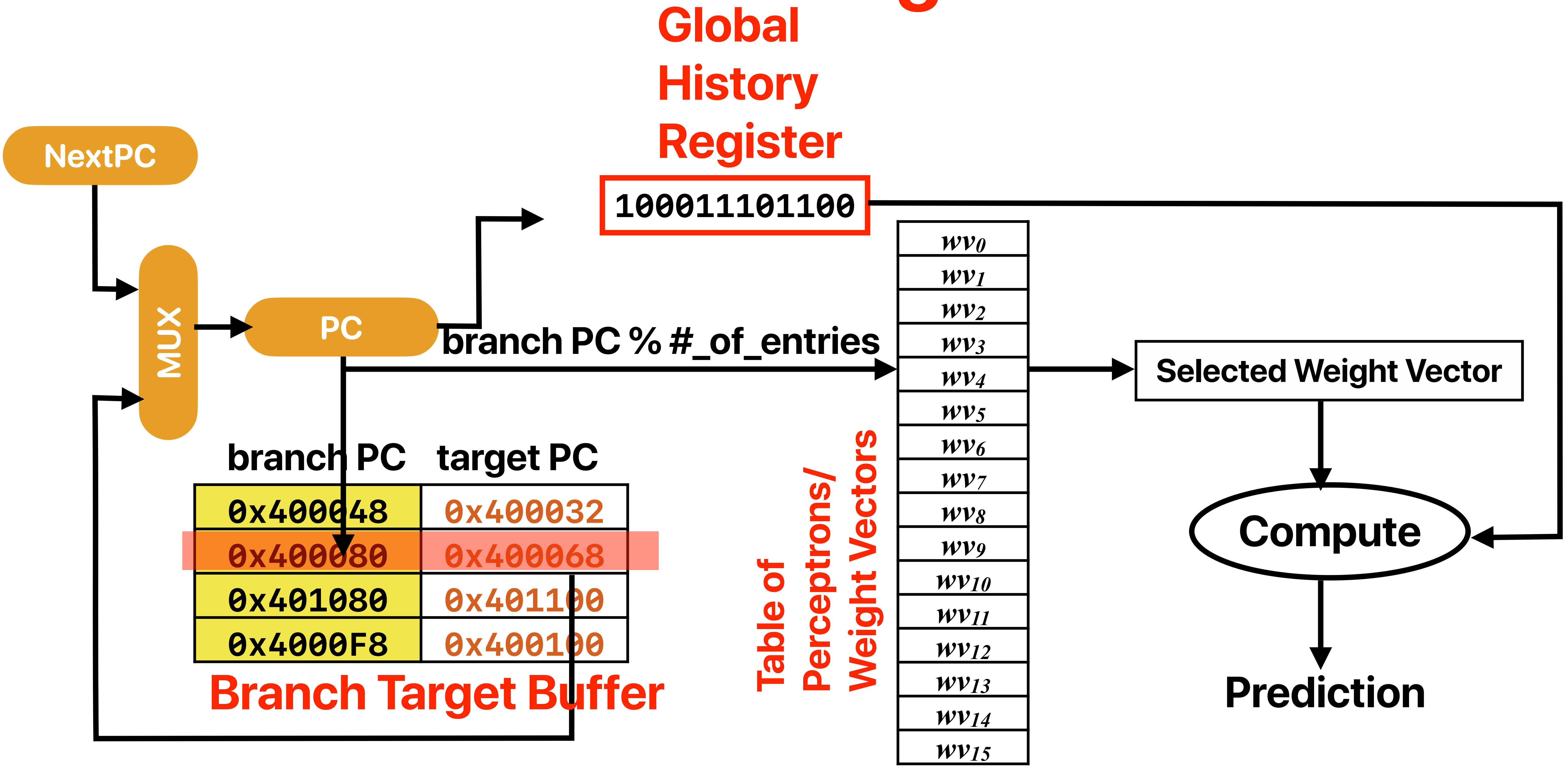
Mapping Branch Prediction to NN (cont.)

- Inputs (x's) are from branch history and are -1 or +1
- n + 1 small integer weights (w's) learned by on-line training
- Output (y) is dot product of x's and w's; predict taken if y = 0
- Training finds correlations between history and outcome



$$y = w_0 + \sum_{i=1}^{n} x_i w_i$$

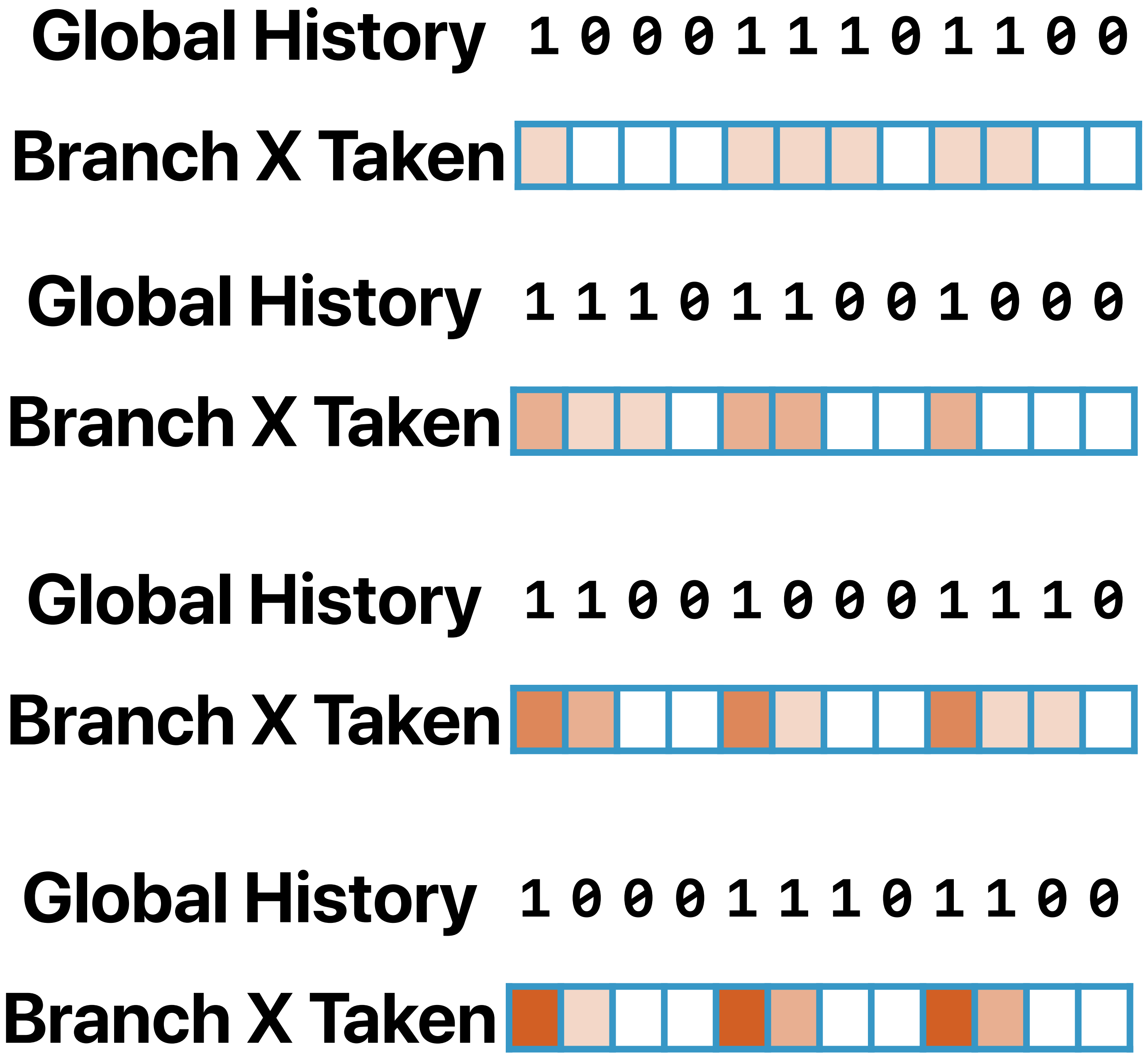
Predictor Organization



Training Algorithm

$x_{1..n}$ is the n-bit history register, x_0 is 1.
$w_{0..n}$ is the weights vector.
t is the Boolean branch outcome.
θ is the training threshold.

if $|y| \leq \theta$ **or** $((y \geq 0) \neq t)$ **then**
    for each $0 \leq i \leq n$ **in parallel**
        if $t = x_i$ **then**
            $w_i := w_i + 1$
        else
            $w_i := w_i - 1$
        end if
    end for
end if



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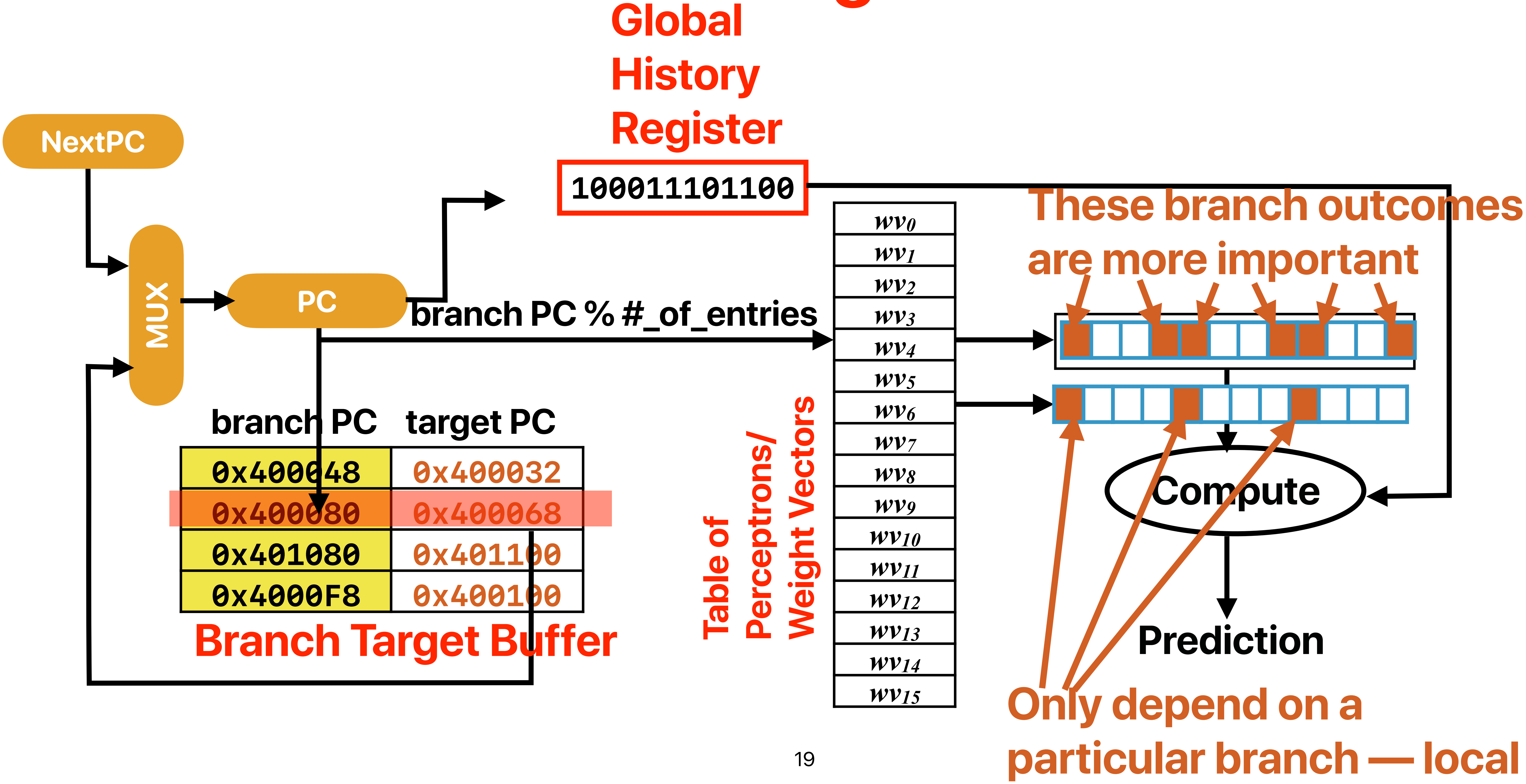

TAGE

André Seznec. The L-TAGE branch predictor. Journal of Instruction Level Parallelism (http://wwwjilp.org/vol9), May 2007.

Better predictor?

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II — about the same

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i = 0;
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III

```
i = 0;
do {
    j = 0;
    do {
      sum += A[i*2+j];
    }
    while( ++j < 2);
} while ( ++i < 100);
```

IV — L could be better

```
i = 0;
do {
    if( rand() %2 == 0)
       a[i] *= 2;
    a[i] += i;
} while ( ++i < 100)
```

A. 0

B. 1

C. 2

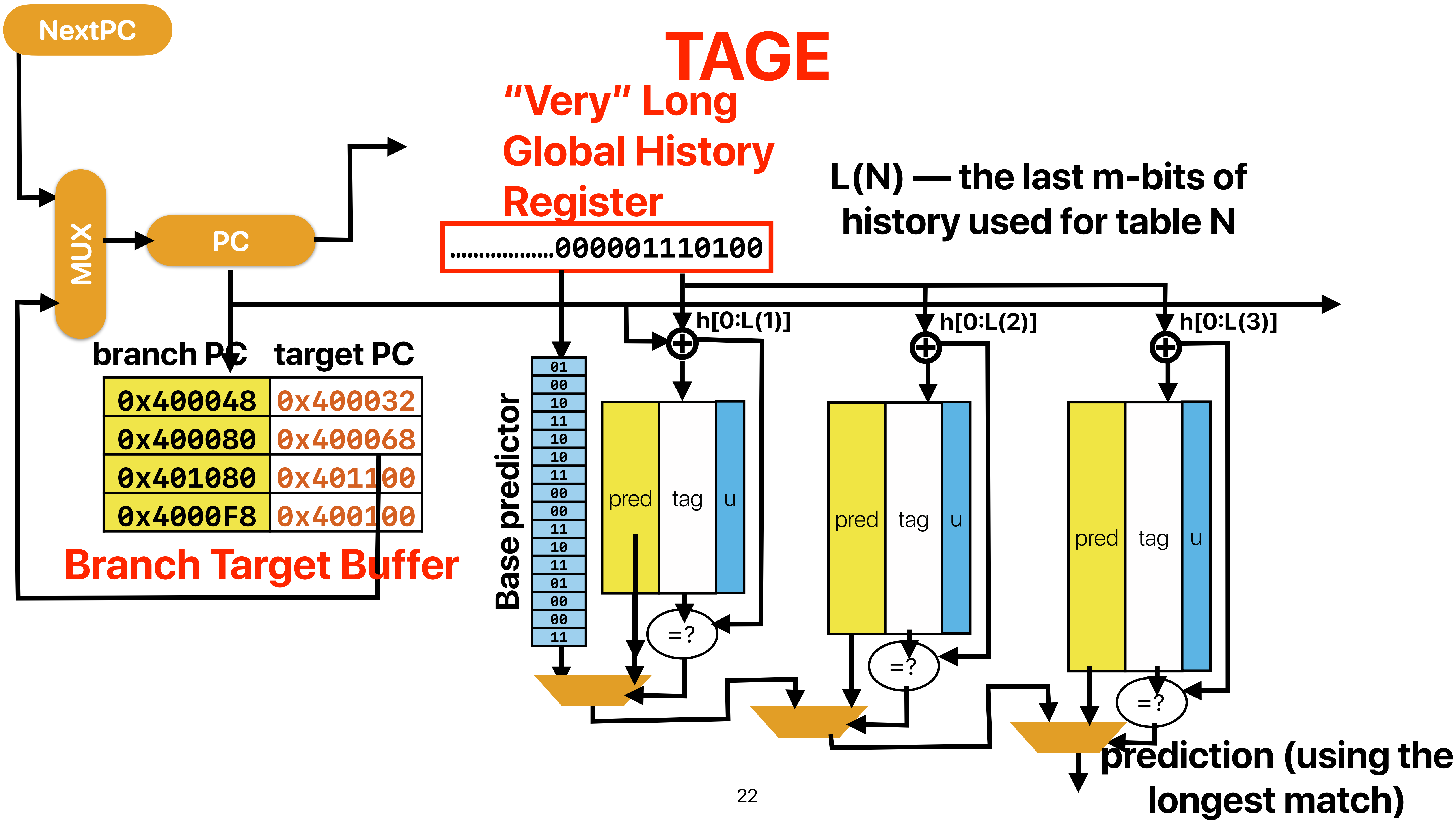
D. 3

E. 4

different branch needs different length of history

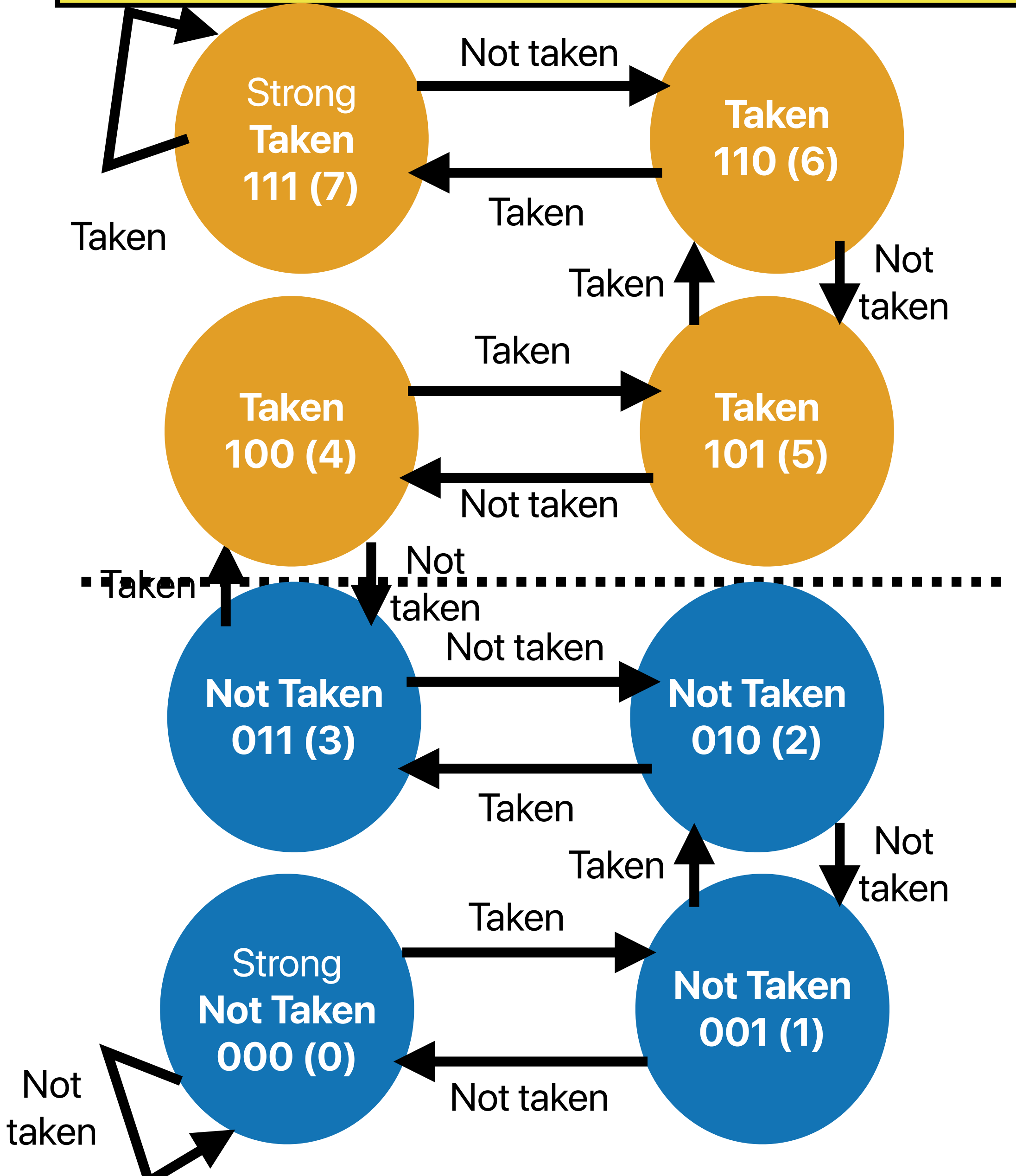
global predictor can work if the history is long enough!

TAGE


NextPC
MUX
PC
"Very" Long
Global History
Register
..................000001110100
L(N) — the last m-bits of
history used for table N
h[0:L(1)]
h[0:L(2)]
h[0:L(3)]
branch PC
target PC
0x400048 0x400032
0x400080 0x400068
0x401080 0x401100
0x4000F8 0x400100
Branch Target Buffer
Base predictor
01
00
10
11
10
10
11
00
00
11
10
11
01
00
00
11
pred
tag
u
pred
tag
u
pred
tag
u
=?
=?
=?
prediction (using the
longest match)

What's inside each table?

pred (3-bit counter)	tag (partial branch PC)	u (usefulness)

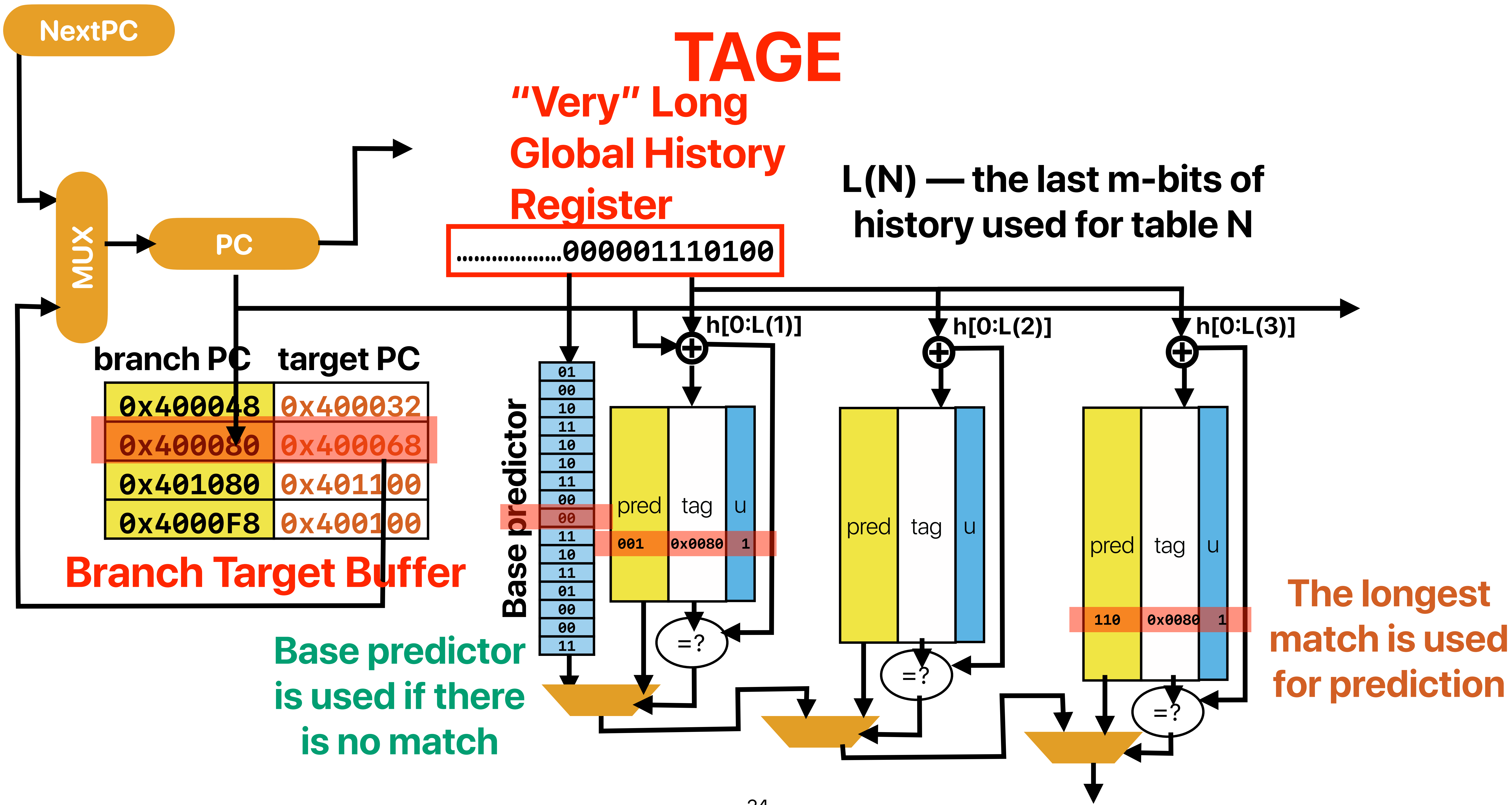


$$\text{if } prediction(alt\_predictor) \neq prediction(pred):$$

$$\text{if } prediction(pred) = actual\ result : u = u + 1$$

$$\text{if } prediction(pred) \neq actual\ result : u = u - 1$$

TAGE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Branch Target Buffer
Base predictor
01
00
10
11
10
10
11
00
00
11
10
11
01
00
00
11
pred
tag
u
001
0x0080
1
pred
tag
u
pred
tag
u
110
0x0080
1
=?
=?
=?
Base predictor
is used if there
is no match
The longest
match is used
for prediction

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Design decisions in real practice

- AMD Zen 2 (RyZen 3000 series processors) adopts a design with first level predictor using perceptron and using TAGE for the 2nd level. What such a design decision implies about the characteristics of TAGE and Perceptron?
  - ① Perceptron takes longer to train than TAGE
  - ② Perceptron takes longer to predict than TAGE
  - ③ Perceptron is more accurate than TAGE
  - ④ Perceptron's performance improves less given more area

  A. 0

  B. 1

  C. 2

  D. 3

  E. 4

> perceptron predictors. For this reason, TAGE was a good choice for [redacted] L2 predictor while keeping perceptron as the L1 predictor for [redacted].

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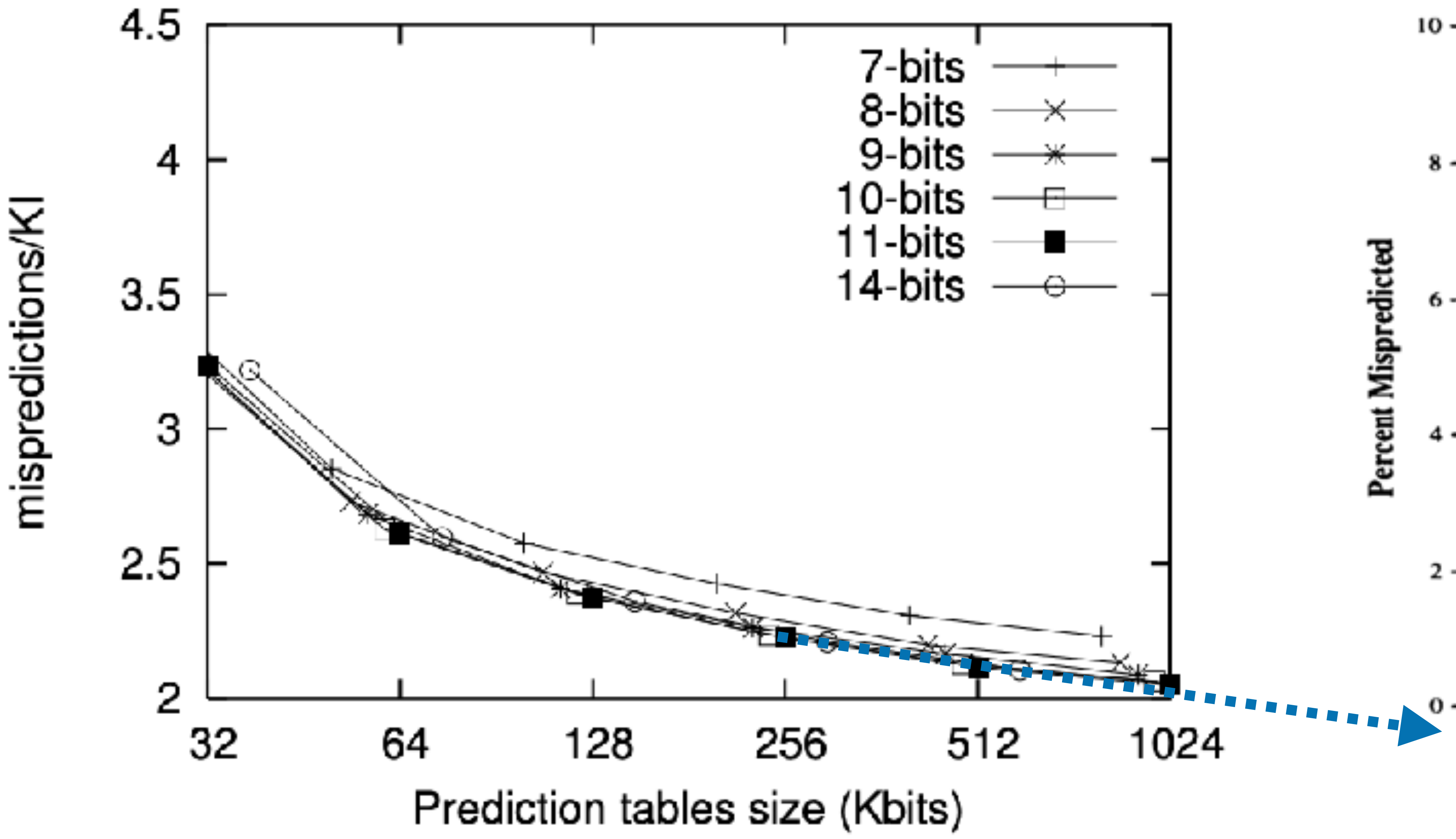
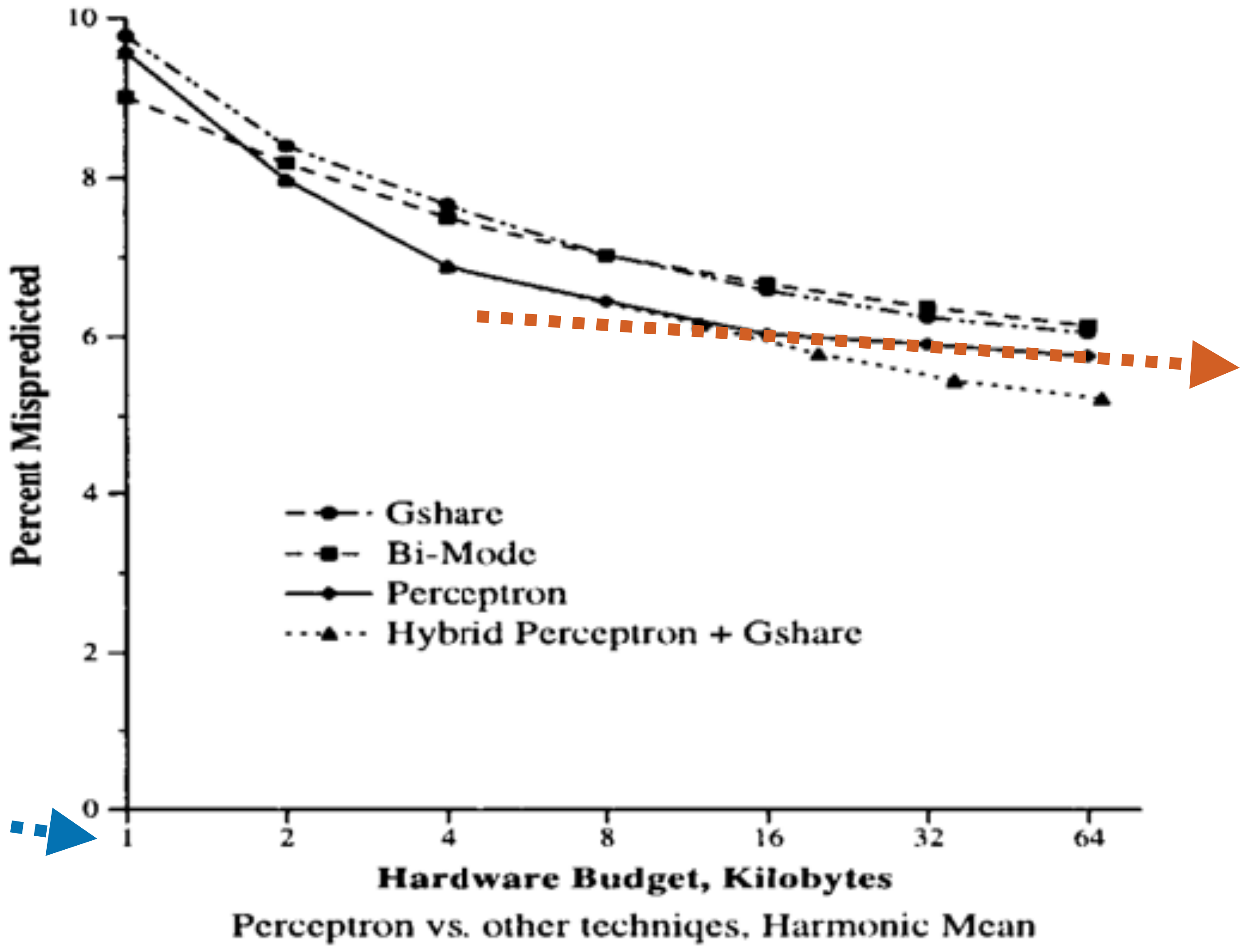
Area efficiency between TAGE and Perceptron



TAGE



Perceptron

How good is prediction using perceptrons?

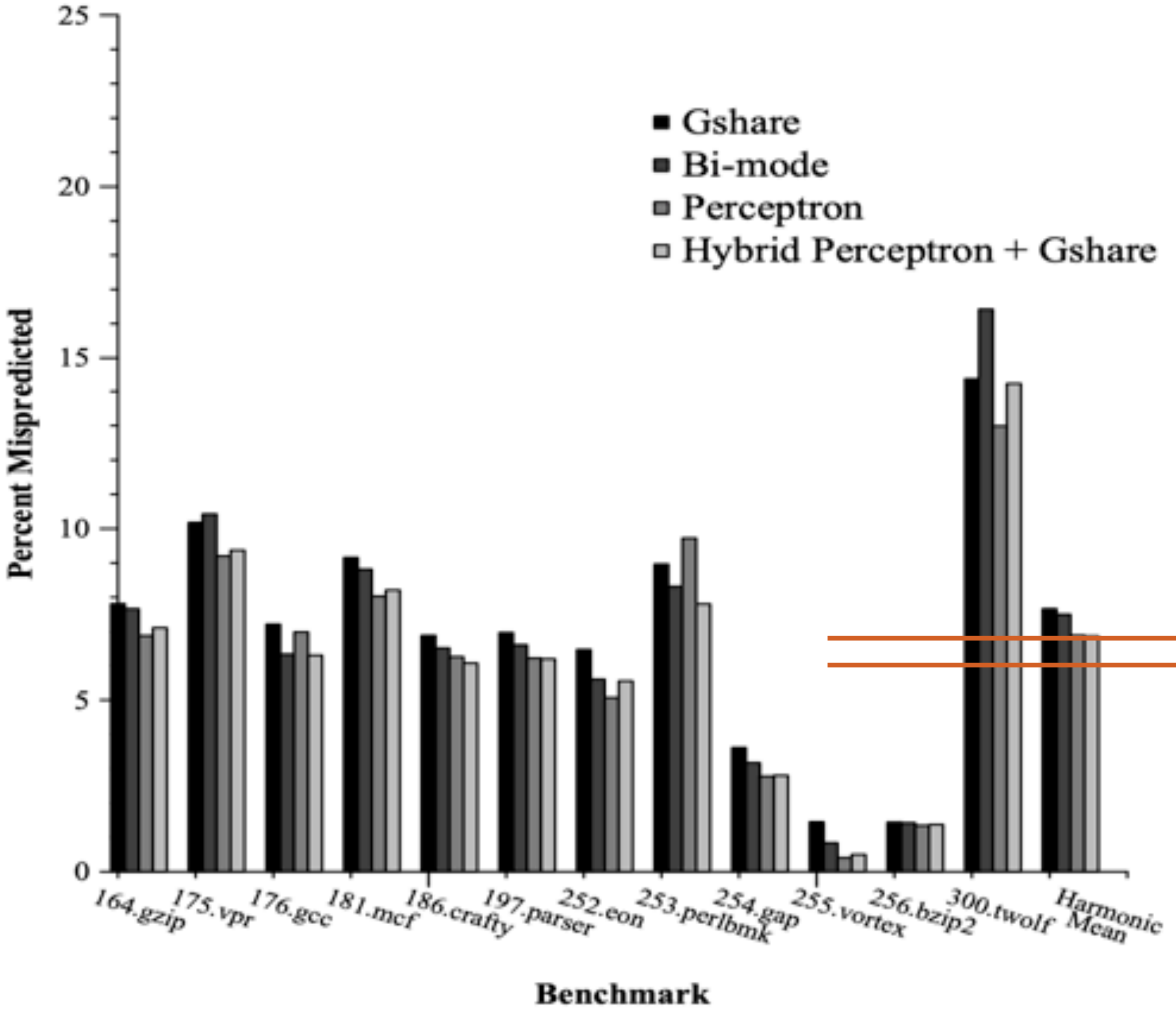


Figure 4: Misprediction Rates at a 4K budget. The perceptron predictor has a lower misprediction rate than *gshare* for all benchmarks except for `186.crafty` and `197.parser`.

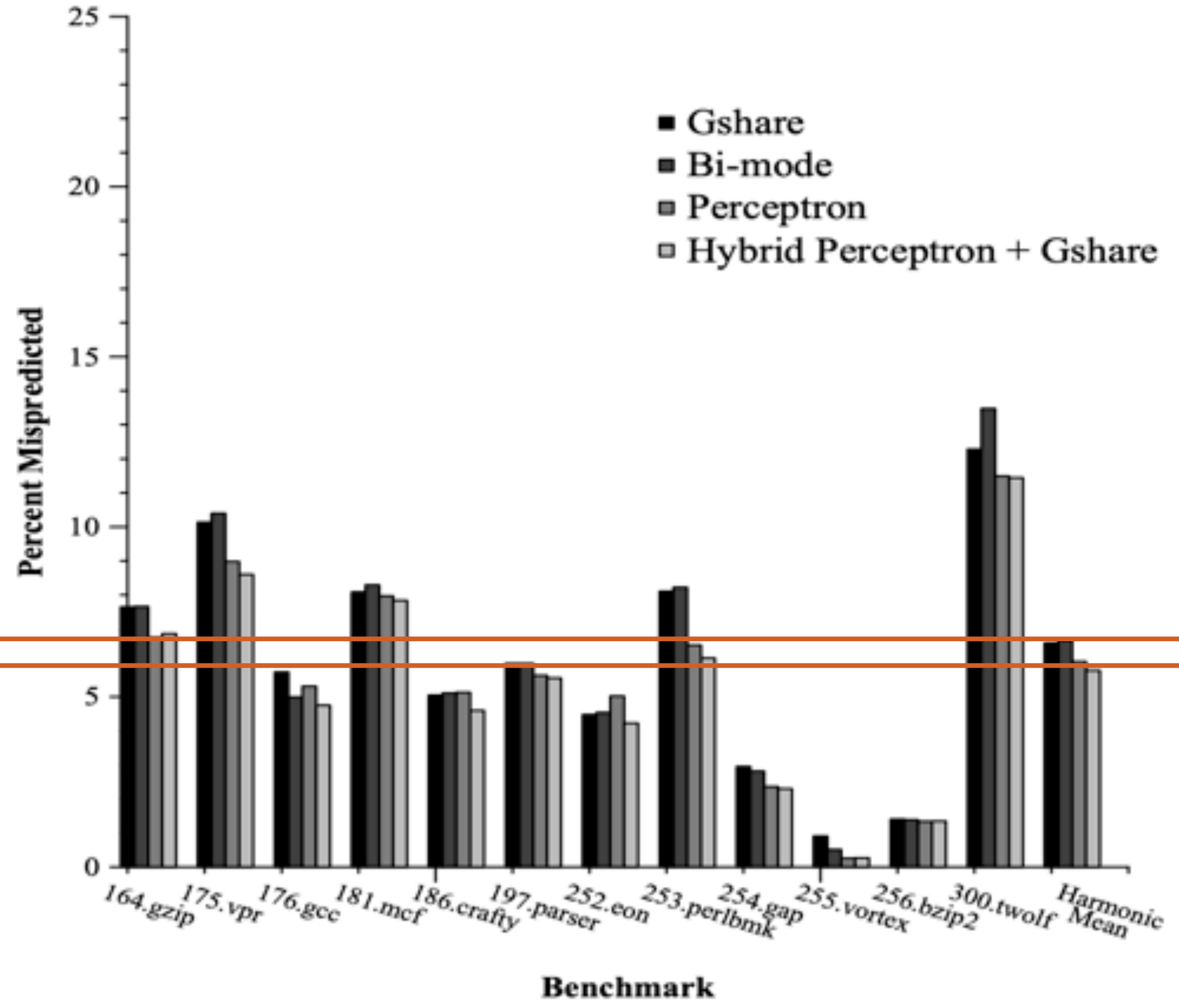
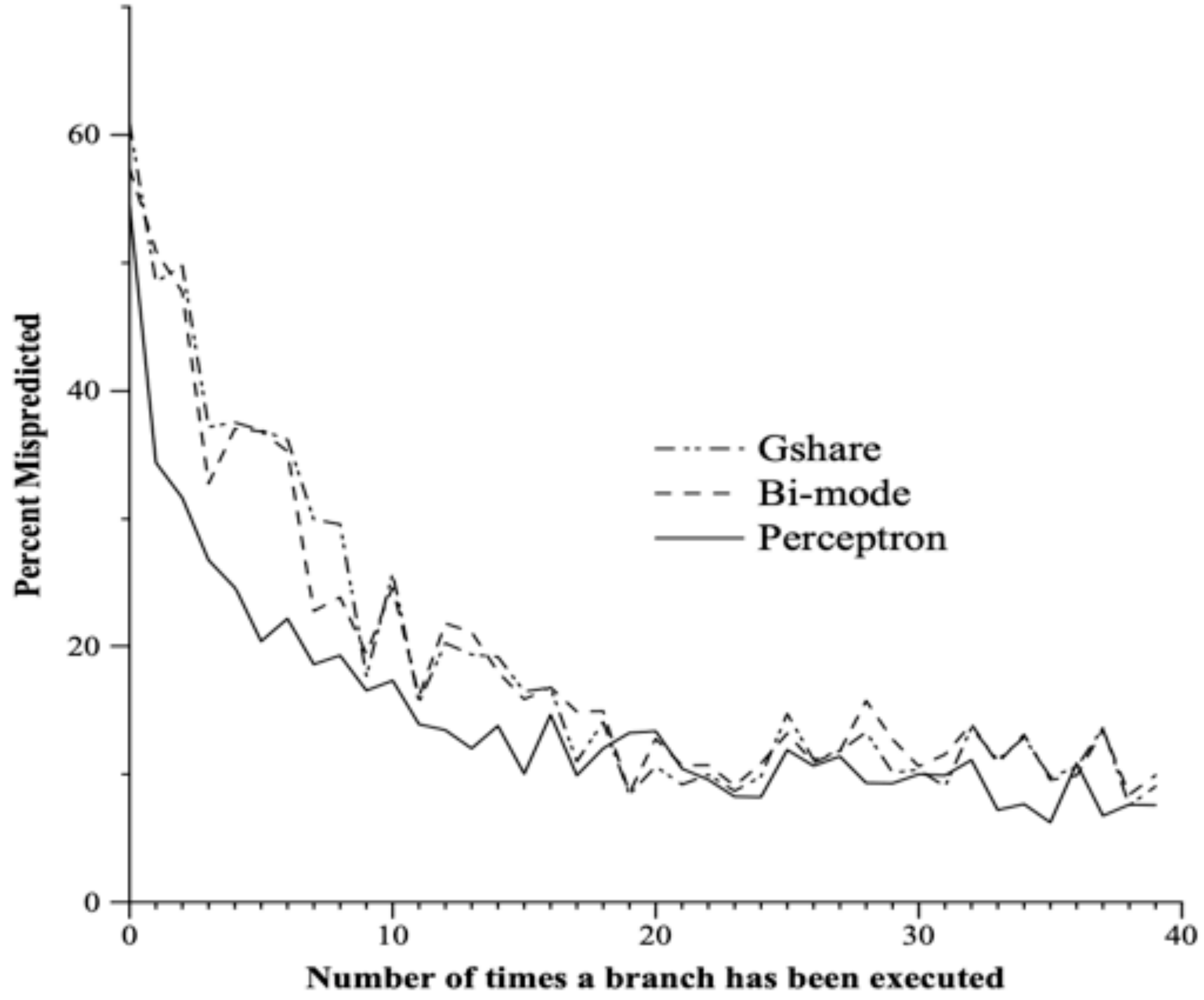


Figure 5: Misprediction Rates at a 16K budget. Gshare outperforms the perceptron predictor only on `186.crafty`. The hybrid predictor is consistently better than the PHT schemes.

History/training for perceptrons



Hardware budget in kilobytes	History Length		
	gshare	bi-mode	perceptron
1	6	7	12
2	8	9	22
4	8	11	28
8	11	13	34
16	14	14	36
32	15	15	59
64	15	16	59
128	16	17	62
256	17	17	62
512	18	19	62

Table 1: Best History Lengths. This table shows the best amount of global history to keep for each of the branch prediction schemes.

AMD Zen 2's design experience

PREDICTION, FETCH, AND DECODE

The in-order front-end of the Zen 2 core includes branch prediction, instruction fetch, and decode. The branch predictor in Zen 2 features a two-level conditional branch predictor. To increase prediction accuracy, the L2 predictor has been upgraded from a perceptron predictor in Zen to a tagged geometric history length (TAGE) predictor in Zen 2.[5] TAGE predictors provide high accuracy per bit of storage capacity. However, they do multiplex read data from multiple tables, requiring a timing tradeoff versus perceptron predictors. For this reason, TAGE was a good choice for the longer-latency L2 predictor while keeping perceptron as the L1 predictor for best timing at low latency.

D. Suggs D. Bouvier M. Subramony and K. Lepak "Zen 2" Hot Chips vol. 31 2019.

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- AMD Zen 2 (RyZen 3000 series processors) adopts a design with first level predictor using perceptron and using TAGE for the 2nd level. What such a design decision implies about the characteristics of TAGE and Perceptron?

① Perceptron takes longer to train than TAGE

no — based on the paper

② Perceptron takes longer to predict than TAGE

short — otherwise won't be in L1

③ Perceptron is more accurate than TAGE

less accurate — otherwise won't need an L2

④ Perceptron's performance improves less given more area

no — based on the paper

A. $\boxed{0}$

B. 1

C. 2

D. 3

E. 4

PREDICTION, FETCH, AND DECODE

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Branch predictors in processors

- The Intel Pentium MMX, Pentium II, and Pentium III have local branch predictors with a local 4-bit history and a local pattern history table with 16 entries for each conditional jump.
- Global branch prediction is used in Intel Pentium M, Core, Core 2, and Silvermont-based Atom processors.
- Tournament predictor is used in DEC Alpha, AMD Athlon processors
- The AMD Ryzen multi-core processor's Infinity Fabric and the Samsung Exynos processor include a perceptron based neural branch predictor.

Takeaways: branch predictions

- The cost of not to predict a branch is to stall until the data dependency is resolved — 34 cycles on modern intel processors and 23 on AMD processors
- Branch predictions allow the processor to at least make some progress and hide the stalls if we guessed correctly!
- Dynamic branch prediction — predict based on prior history
  - Local predictor — make predictions based on the state of each branch instruction
  - Global predictor — make predictions based on the state from all branches
  - Both are not perfect — hybrid predictors
    - Tournament
    - gshare
    - Perceptron
    - TAGE
  - All modern processors have pretty accurate branch predictors — if the code itself is predictable

Recap: But A is faster!

d. /* one line statement using bit-wise operators */ (most efficient)
a^=b^=a^=b;

The order of evaluation is from right to left. This is same as in approach (c) but the three statements are compounded into one statement.

A

```c
void regswap(int* a, int* b) {
    int temp = *a;
    *a = *b;
    *b = temp;
}
```

B

```c
void xorswap(int* a, int* b) {
    *a ^= *b = *a = *b;
}
```

Data hazards

Data hazards

- An instruction currently in the pipeline cannot receive the “logically” correct value for execution
- Data dependencies
  - The output of an instruction is the input of a later instruction
  - **May sometimes** result in data hazard if the later instruction that consumes the result is still in the pipeline

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How many data dependencies do we have?

- How many pairs of data dependences are there in the following x86 instructions?

```
movl    (%rdi), %eax
movl    (%rsi), %edx
movl    %edx, (%rdi)
movl    %eax, (%rsi)
```

```c
int temp = *a;
*a = *b;
*b = temp;
```

A. 1
B. 2
C. 3
D. 4
E. 5

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How many data dependencies do we have?

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```
movl    (%rdi), %eax
xorl    (%rsi), %eax
movl    %eax, (%rdi)
xorl    (%rsi), %eax
movl    %eax, (%rsi)
xorl    %eax, (%rdi)
```

```
*a ^= *b;
*b ^= *a;
*a ^= *b;
```

A. 1
B. 2
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xorl     (%rsi), %eax
movl     %eax, (%rdi)
xorl     (%rsi), %eax
movl     %eax, (%rsi)
xorl     %eax, (%rdi)
```

```
*a ^= *b;
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A. 1
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Data hazards

```
① movl    (%rdi), %eax
② movl    (%rsi), %edx
③ movl    %edx, (%rdi)
④ movl    %eax, (%rsi)
```

	IF	ID	ALU/BR/AG	M1	M2	M3	M4/XORL	WB/Retire
1	(1)							
2	(2)	(1)						
3	(3)	(2)	(1)					
4	(4)	(3)	(2)	(1)				
5		(4)	(3)	(2)	(1)			
6			(4)	(3)	(2)	(1)		
7				(4)	(3)	(2)	(1)	
8					(4)	(3)	(2)	
9						(4)	(3)	
10							(4)	
11								
12								
13								
14								

%edx does not have our desired value

%eax does not have our desired value

Solution 1: Let’s try “stall” again

- Whenever the input is not ready when the consumer is decoding, just stall — the consumer stays at ID.

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Data hazards?

- How many cycles do we have to stall in the following x86 instructions to get the expected output if a memory operation (assume 100% cache hit rate) takes 5 cycles?

```
① movl    (%rdi), %eax
② movl    (%rsi), %edx
③ movl    %edx, (%rdi)
④ movl    %eax, (%rsi)
```

A. 1
B. 2
C. 3
D. 4
E. 5

Data hazards?

- How many cycles do we have to stall in the following x86 instructions to get the expected output if a memory operation (assume 100% cache hit rate) takes 5 cycles?

```
① movl    (%rdi), %eax
② movl    (%rsi), %edx
③ movl    %edx, (%rdi)
④ movl    %eax, (%rsi)
```

A. 1
B. 2
C. 3
D. 4
E. 5

	IF	ID	ALU/BR/AG	M1	M2	M3	M4/XORL	WB/Retire
1	(1)							
2	(2)	(1)						
3	(3)	(2)	(1)					
4	(4)	(3)	(2)	(1)				
5	(4)	(3)		(2)	(1)			
6	(4)	(3)			(2)	(1)		
7	(4)	(3)				(2)	(1)	
8	(4)	(3)					(2)	(1)
9	(4)	(3)						(2)
10		(4)	(3)					
11			(4)	(3)				
12				(4)	(3)			
13					(4)	(3)		
14						(4)		(3)
15								(4)

5 cycles stalls

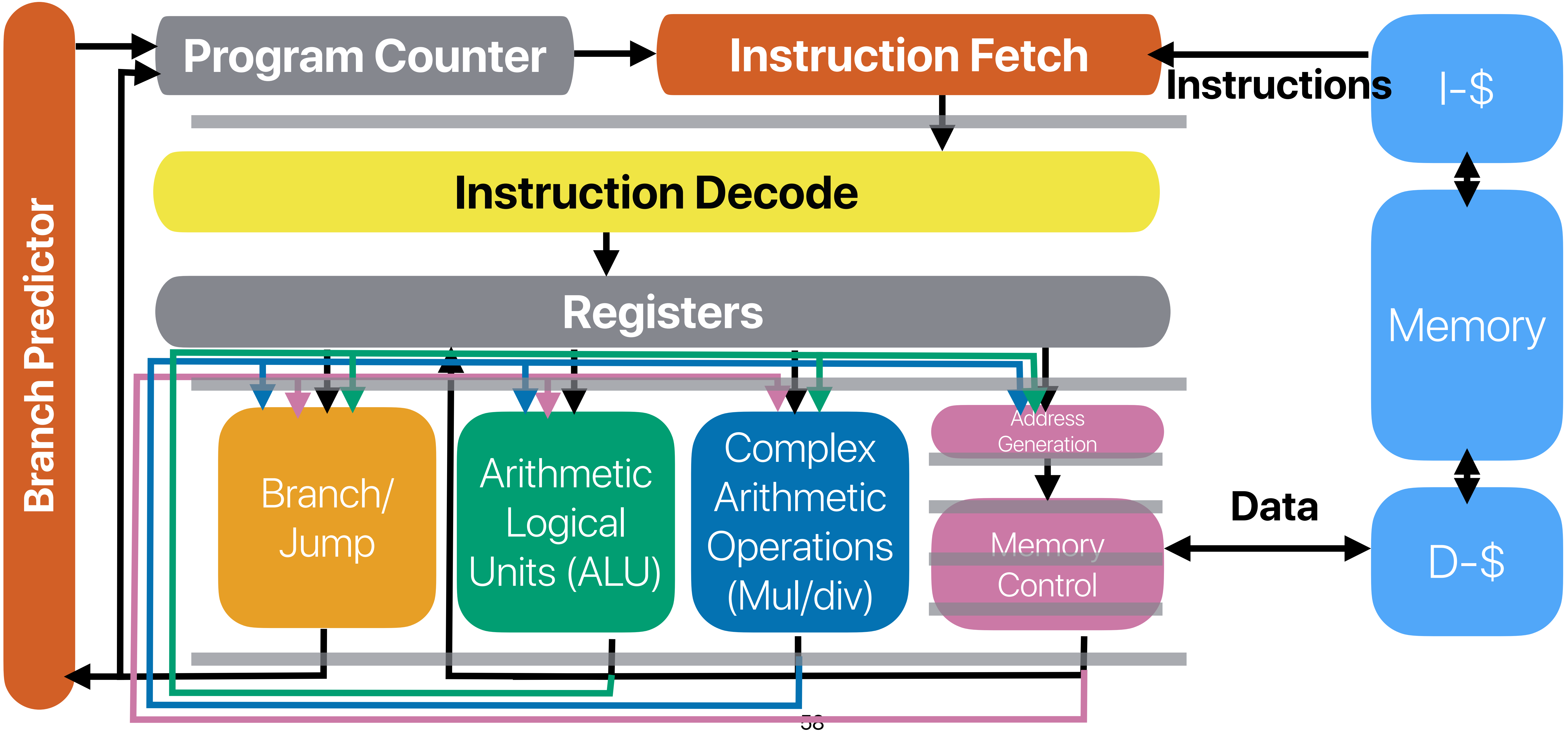
we have the value for %edx already!

Why another cycle?

Solution 2: Data forwarding

- Add logics/wires to forward the desired values to the demanding instructions

Data "forwarding"



The effect of data forwarding

```
① movl    (%rdi), %eax
② movl    (%rsi), %edx
③ movl    %edx, (%rdi)
④ movl    %eax, (%rsi)
```

	IF	ID	ALU/BR/AG	M1	M2	M3	M4/XORL	WB/Retire
1	(1)							
2	(2)	(1)						
3	(3)	(2)	(1)					
4	(4)	(3)	(2)	(1)				
5	(4)	(3)		(2)	(1)			
6	(4)	(3)			(2)	(1)		
7	(4)	(3)				(2)	(1)	
8	(4)	(3)					(2)	(1)
9		(4)	(3)					(2)
10			(4)	(3)				
11				(4)	(3)			
12					(4)	(3)		
13						(4)	(3)	
14							(4)	(3)
15								(4)

4 cycles stalls

8 cycles for 4 instructions
CPI = 2

Announcements

- **Programming assignment 2** due this evening
- **Reading Quiz 7** due next **Thursday** before the lecture
- Assignment 4 is already up and due 11/20

Computer
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Engineering

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